

## Module 3

### Operational Vs Decisional Support System

| Operational (OSS)                                               | Decisional (DSS)                                                                   |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------|
| In terms of purpose, OSS automates routine tasks and processes. | In terms of purpose, DSS provides data and tools for decision making and analysis. |
| In terms of data, OSS has structural and transactional data.    | In terms of data, DSS has historical and predictive data.                          |
| OSS supports transactions which are happening in real-time.     | DSS supports transactions which have happened a long time ago to analyze and make. |
| The concurrent transaction volume is very high.                 | Concurrent transaction levels in DSS are medium or low.                            |
| They are regarded as update transactions.                       | They are regarded as query transactions.                                           |

### Data Warehousing:

Data Warehousing is a process of collecting, organizing and storing large amounts of data from various sources in a centralized repository. It is designed to support business intelligence activities such as data analysis, reporting and decision making.

### Features of Data warehousing:

- **Subject-oriented:** A data warehouse is designed around specific subjects or topics related to an organization's business such as sales, customer data and financial data.
- **Integrated:** Data in a data warehouse is integrated from multiple sources including internal systems and external data sources. It helps data stay accurate.
- **Time Variant:** Data in a data warehouse is stored over time allowing analysts to track changes in data and identify trends over time.
- **Non- Volatile:** Data in a data warehouse is read-only, so data cannot be updated or changed directly.

### Need for Data Warehousing:

- It is difficult to retrieve data from tables for analytical purposes.
- Data warehousing is the best way to integrate valuable data from different sources into the database of a particular application.
- Data warehouses make it easy to develop and store metadata.
- Business experts or users become habitual to see many customized data on display screens fields such as rolled-up general ledger balances. These fields do not exist in the database.

- When we perform reporting and analysis functions on the hardware that handles transactions, the performance is often poor.

Therefore, data warehouses should be used for reporting and analysis.

**Data Warehouse Architecture:**

**Data Source Layer:** Refers to the layer representing various data sources that enter data into the data warehouse. The data can be in any of these formats: Plain text file, relational database, excel file and other types of databases. The following are the types of data that can act as data sources:

- Production Sources: Represents sales data, HR data and product data.
- Internal Data: Represents data of a department or an organization such as employee data
- External Data : Represents data from outside the organization or third party data such as census data or demographic or survey data.
- Archived data: Represents logs of the Web server along with the user's browsing data

**Data Staging Layer:** Refers to the storage area for data processing where data comes before being transformed into the data that is entered in a data warehouse. The following are the steps involved in transporting data from various sources to data warehouse:

- Extraction: Refers to the process of extracting data from different source systems and validating it against certain quality.
- Transformation: Refers to the transformation of the data available in different source systems and validating it against certain quality.
- Loading: Refers to the process of loading the data either from data warehouse or data mart.

**Data Storage Layer:** Refers to the layer in which the transformed data and clean data is stored. On the basis of the scope and functionality. The following are the types of entities in this stage.

- Data warehouse : It is maintained by organizations as a central warehouse of data that can be equally accessed by all business experts and end users.
- Data Mart : When a data warehouse is created at the departmental level. It is known as data mart.
- Meta Data: Details about the data is known as metadata. In other words, it is a catalog of data warehouses.
- MDDb: It is a multidimensional database that allows data to be molded and viewed in multiple dimensions. It is defined by dimensions and facts.
- Information Delivery: Provide information that reaches end users.

### **Information Flow Mechanism:**

- Source Data
- Data Staging Area
- Data Warehouse
- Data Warehouse User

### **Transformation of data into information:**

- Select the source data
- Extract the data from the source system
- Transform the extracted data
- Load the transformed data into the data warehouse
- Deliver the information to the end users

This process is also called the ETL process that is Extract - Transform - Load

### **Information Flow Architecture:**

- Operational data store (ODS): ODS is a database which stores the latest and real time data from various operating systems.
- ETL Process:
  - Extract: Data is extracted from ODS and other source systems such as CRM, ERP.

- Transform: The extracted data is then transformed into a format that is suitable for the data warehouse
- Load: Transformed data is then loaded onto the data warehouse. This involves putting data into appropriate tables and columns
- Analysis Store: Data is organized, structured and stored in a way which makes it easy to query and analyze.
- OLAP (On-Line Analytical Processing): Organized data is a multidimensional structure. This allows users to analyze data from different perspectives. OLAP provides tools to analyze trend patterns and anomalies.
- Visualization: Creates reports, patterns, trends, insights, dashboards, etc using data within the data warehouse.

### **Meta data:**

Metadata refers to the information about the data stored in the warehouse. It provides a description of data which is stored in the databases such as origin, format and structure.

Metadata plays a key role in data warehousing by providing valuable information. This helps in:

- Understanding the data: Helps users to analyze and report by knowing the origin, format and structure.
- Managing data quality: Metadata also includes information regarding the accuracy, precision, and completeness of the information.
- Data information: Metadata helps integrate data from multiple sources and ensures data is consistent across the organization.
- Data Governance: Metadata plays an important role in data governance by providing information regarding data ownership, data usage, and data security
- Data Lineage : Metadata provides information about the history of data which helps the users trace the lineage and understand concepts.

### **Classification of Metadata:**

- Technical Metadata: Gives information regarding the technical details like format of data, storage, data model, data dictionaries, etc.
- Operational Metadata: Operational metadata gives information regarding data integration, data transformation, data loading and data quality.
- Business Metadata: Contains business concepts represented by data.
- Usage Metadata: Contains information regarding query execution, report generation, and data visualization used to analyze data.
- Data Lineage Metadata: Describes the history of the data transformation, date it came into the warehouse, etc.

### Data Warehouse vs Data Marts:

| Data Warehouse                                                                                            | Data Mart                                                                                              |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Data warehouses have a vast repository of collected information for internal as well as external sources. | Data mart is a subcategory of data warehouses created to meet the requirements of a smaller sub-group. |
| Data warehouse holds information related to multiple topics and subjects within an organization           | Data marts hold information related to a specific domain or a single subject within an organization    |
| It has detailed information about every subject                                                           | It has summarized information regarding a single domain                                                |
| All the sources of information are integrated                                                             | The single source containing the specific data is integrated.                                          |
| Data warehouse has a centralized system.                                                                  | Data mart has a decentralized/distributed system.                                                      |

### Data Warehousing Design Strategies:

Data warehouses are used across multiple businesses to help with critical decision making and data analysis.

It has two main approaches called Top-Down approach and Bottom-up approach:

**Top-Down Approach:** Also known as enterprise data warehouse. (EDW) which involves creating a comprehensive, centralized data warehouse that serves an entire organization.

Advantages Of Top-Down Approach:

- Ensure data consistency and accuracy
- Developing data marts is easier

Disadvantages of Top-Down Approach:

- Inflexible to change in departmental needs
- Cost is high

**Bottom- Up Approach:**

Also known as the data mart approach. This approach involves small targeted data marts that serve specific business functions and departments.

Advantages of Bottom-Up Approach:

- Documents are generated quickly.
- Cheaper than Top-Down approach

Disadvantages of Bottom-Up Approach:

- Leads to data inconsistency
- Difficult to manage and govern data.

## Data Warehouse Modeling vs Operational Database Modeling

| <b>Data Warehouse Modelling</b>                       | <b>Operational Database Modeling</b>          |
|-------------------------------------------------------|-----------------------------------------------|
| Operates on schemas like star, snowflake schema, etc. | Operates on normalized data modeling          |
| Emphasis on data quality and data consistency         | Emphasis on data accuracy                     |
| Designed for read intensive operations                | Designed for both read and write operations   |
| Designed to manage large volumes of data              | Designed to manage smaller but real time data |

**Fact Table:** A fact table or a fact entity is a table or entity in a star or snowflake schema that stores measures that measure the business, such as sales, cost of goods, or profit. Fact tables and entities aggregate measures, or the numerical data of a business.

**Factless Fact table:** Factless fact tables are only used to establish relationships between elements of different dimensions

**Dimension Table:** A dimension table or dimension entity is a table or entity in a star, snowflake, or starflake schema that stores details about the facts. For example, a Time dimension table stores the various aspects of time such as year, quarter, month, and day.

### **Star Schema:**

In star schema, data is organized into a central fact table that contains the measures of interests, surrounded by dimension tables that describe the attributes of the measure

In star schema, quantitative data is stored in fact tables and dimensions which are the context for the quantitative data is stored in dimension tables.

Each dimension table is joined to the fact table through a foreign key relationship. This allows query executions in the fact table using attributes from the dimension table.

**Advantages of Star Schema:**

- Simpler queries
- Simplified business reporting logic
- Feeding cubes in OLAP

**Disadvantages of Star Schema:**

- Highly denormalized
- Does Not reinforce many-to-many relationships within business entities

**Snowflake Schema:**

Snowflake schema is the extended version of the star schema where the centralized fact table is connected to multiple dimensions. There are several levels of relationships and hierarchies.

The fact table is located at the center of the schema surrounded by dimension tables. Each dimension table is broken down into smaller dimensions.

**Advantage:**

- Reduces data redundancy and improves consistency
- Improves data integrity

**Disadvantages:**

- Complex to prepare
- Slower response time

**Fact Constellation Schema:**

It is a collection of multiple star schemas where multiple fact tables have shared dimensions.

### **Types of Keys:**

- **Primary Key** – Every row in a dimension table is identified by a unique value which is generally known as primary key. The primary key is a unique identifier that helps isolate each row in a dimension. Tables can have numerous records, however each record has only one Primary Key.
- **Surrogate Key** – These are the keys which are generated by the system and generally do not have any built in meaning. It is UNIQUE and SEQUENTIAL since it is a consecutively created number for each record in the table. It is MEANINGLESS since it doesn't have any business significance other than identifying each row. For example, if a data warehouse contains information on 20,000 clients the dimension table will contain 20,000 surrogate keys one for each client.
- **Foreign Key** – In the fact table the primary key of another dimension table acts as the foreign key.

### **Slowly Changing Dimensions (SCD):**

As data constantly increases due to high transaction rates, the fact tables tend to change frequently over time changes can occur in dimension tables due to change in their attributes. The types of SCDs are Type-1, Type-2, Type -3 SCD.

**Type-1 SCD:** Overwriting existing data having no record or log of the past and no changes except overwriting.

**Type-2 SCD:** Tracking all the changes that are made by creating a new row for updated values

**Type-3 SCD:** The change in creating new fields in the dimension record to capture the last value for that attribute.

**Rapidly Changing Dimensions:**

When one or more attributes change frequently.

Eg. BMI, Height, Weight

If any of these attributes change rapidly, many rows are created and the dataset would get larger and larger for minute changes. These attributes are removed and put in a separate dimension called Junk dimension.

**Junk Dimensions:**

A junk dimension is a convenient grouping of typically low-cardinality flags and indicators. By creating an abstract dimension, these flags and indicators are removed from the fact table while placing them into a useful dimensional framework.

**Aggregate Fact Table:**

Aggregate fact tables are simple numeric rollups of atomic fact table data built solely to accelerate query performance. These aggregate fact tables should be available to the BI layer at the same time as the atomic fact tables so that BI tools smoothly choose the appropriate aggregate level at query time.

**Data Lake:**

A data lake is a large, centralized repository that allows you to store and manage vast amounts of raw data in its native format. Unlike traditional data warehouses, data lake can store data in semi-structured, unstructured and structured format.

Architecture:

## Data Warehouse vs Data Lake

| Parameters      | Data Lake                                                                                                  | Data Warehouse                                              |
|-----------------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Data type       | Raw (all types, no matter source of structure)                                                             | Processed (data stored according to metrics and attributes) |
| Data purpose    | To be determined                                                                                           | Currently being used                                        |
| Process         | Extract Load Transform (ELT)                                                                               | Extract Transform Load (ETL)                                |
| Schema position | After data storage, to offer agility and easy data capture                                                 | Before data storage, to offer security and high performance |
| Users           | Data scientists, those who need in-depth analysis and tools (such as predictive modeling) to understand it | Business professionals, those who need it for operations    |
| Accessibility   | Accessible and easy to update                                                                              | Complicated to make changes                                 |
| History         | Relatively new for big data                                                                                | The concept has been around for decades                     |